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Dr. Olga Soloveva

Postdoctoral researcher

Research interests: [hep-ph, nucl-th, hep-lat]
- Phase transitions in strongly correlated systems: in QCD at high temperature and density
- Phenomenology of heavy-ion collisions: thermodynamics and transport phenomena at finite temperature and density
- Hard probes: Jets/heavy quarks in the strongly interacting quark-gluon plasma (sQGP)
- Machine learning application

Current position

Jan 2022 - present **Postdoctoral researcher**, *GSI Helmholtzzentrum für Schwerionenforschung GmbH*, Darmstadt, Germany.
Supervisor: Prof. H. Elfner

Profiles

- Full Name: Olga Evgenyevna Soloveva (Solovjeva in publications 2014-2018)
- ORCID: 0000-0002-8833-4384
- Inspire: [Soloveva-Solovjeva]
- LinkedIn Olga-Soloveva
- gitlab gitlab.com/osolo
- github github.com/osoloque

Education

May 2018 - December 2021 **Ph.D. in Physics (Magna Cum Laude)**, *Goethe University*, Frankfurt am Main, ("Transport properties of the strongly-interacting quark-gluon plasma"); Supervisor: Prof. Dr. E. Bratkovskaya.

September 2014 - August 2016 **Master studies *(Graduation with distinction)***, *National Research Nuclear University MEPhI (Moscow Engineering Physics Institute)*, Moscow, Master's thesis: "Magnetic characteristics of mesons in the lattice gauge theory"; Supervisors: Dr. E.V. Luschevskaya, Prof. Dr. O.V. Teryaev.

September 2010 - August 2014 **Bachelor studies *(Graduation with distinction)***, *National Research Nuclear University MEPhI (Moscow Engineering Physics Institute)*, Moscow, Bachelor's thesis: "Determination of the interaction potential between static charges in graphene within Monte-Carlo methods"; Supervisor: Dr. V.V. Braguta.

September 2000 - August 2010 **Pupil**, *Novosondetskiy Bul'var, 4, Ulyanovsk, Ulyanovsk Oblast, Russia, 432072*, Novosondetskiy Bul'var, 4, Ulyanovsk, Ulyanovsk Oblast, Russia, 432072, email: mou20@uom.mv.ru.

Complementary Education

Apr 2019 **Making an Impact as an effective researcher**, *with the doctoral school HGS-HIRe*, Buchenau, Germany.

Nov 2020 **Leading Teams in a Research Environment**, with the doctoral school HGS-HIRe, (online).

Sep 2022 -present **Machine Learning zoomcamp**, *by Alexey Grigoriev*, (on Zoom) Berlin, Germany.

11-18 Apr 2023 **The Eighth Machine Learning in HEP summer school**, *by INFN*, Erice, Italy.

Funding and Grants

2022 -2024 HFHF Postdoctoral Scholarship

2018 -2021 Ph.D. Scholarship - Helmholtz Graduate School for Hadron and Ion Research (HGS-HIRe)

2016 Edward Lozansky scholarship for students specialized in theoretical physics.

2015 — 2016 Scholarship for students of the federal state educational institutions of professional education in Russian Federation, Government Decree № 945, Nov 18, 2011.

2015 — 2017 Scholarship for young scientists (senior students, Post-graduate students and candidates of sciences) within the framework of the "Student Research and Education Program of the FAIR-Russia Research Center"

Research experience

Jun 2024 - present **Postdoctoral researcher**, *GSI Helmholtzzentrum für Schwerionenforschung GmbH*, Darmstadt, Germany.
AG Prof. Dr. Hannah Elfner

Jan 2022 - Jun 2024 **Postdoctoral researcher**, *Helmholtz Research Academy Hesse for FAIR (HFHF) and ITP, Goethe University*, Frankfurt, Germany.
AG Prof. Dr. Elena Bratkovskaya

2018 - 2021 **Researcher PhD student**, *ITP, Goethe University*, Frankfurt, Germany.
AG Prof. Dr. Elena Bratkovskaya

2016 - 2018 **Junior researcher**, *Institute for Theoretical and Experimental Physics (ITEP)*, Moscow, Russia.
Laboratory 191 "Lattice Gauge Theories"

2015 - 2016 **Junior researcher**, *NRNU MEPhI*, Moscow, Russia.
Department for Theoretical Nuclear Physics

2014 - 2015 **Laboratory assistant**, *Institute for Theoretical and Experimental Physics (ITEP)*, Moscow, Russia.
Laboratory 910 "High Energy Physics"

Service to the Scientific Community

Feb 2023 - present **member of Thematical Working Group 8 (TWG)**, *sub-group Computing, Machine Learning and Artificial Intelligence*, NuPECC Long Range Plan 2024.

2022 - present **referee**, *Physical Review D, Modern Physics Letters A*.

17–29 August 2017 **organizing committee member**, *6th International Conference on New Frontiers in Physics*, Crete, Greece.

Skills

Languages

English	**Fluent**
German	**Upper - Intermediate (B2)**
Italian	**Intermediate (A2)**
Russian	**Native**

Technical skills

Programming languages	Fortran, Python (ML, NN(TensorFlow, Keras), R(basics),LaTeX, Bash, HTML/CSS, Wolfram Mathematica
Operating System	Linux, Windows
HPC	SLURM, Version Control System (git)

Publications

39 publications, 549 citations, h-index: 13 (as of May 9, 2026; source: [INSPIRE])

White paper/ Invited review

- "Phase Transitions in Particle Physics – Results and Perspectives from Lattice Quantum Chromo-Dynamics", (Chapter 3.5), Progress in Particle and Nuclear Physics 104070 (2023) e-print arXiv: 2301.04382 [hep-lat];

Peer-Reviewed Journals

- 1. "$\pi^{\pm}$ and $\rho^{0,\pm}$ - mesons in a strong magnetic field on the lattice"
  E. V. Luschevskaya, O. A. Kochetkov, O. V. Teryaev, **O.E. Solovjeva**,
  JETP Lett. 101 (2015), no. 10, 674-678;
- 2. "Magnetic polarizabilities of light mesons in SU(3) lattice gauge theory"
  E. V. Luschevskaya, **O.E. Solovjeva** O. A. Kochetkov, O. V. Teryaev,
  Nucl. Phys. B **898** (2015) 627-643; e-print arXiv: 1411.4284 [hep-lat];
- 3. "Magnetic polarizability of pion"
  E.V. Luschevskaya, **O.E. Solovjeva**, O.V. Teryaev,
  Phys. Lett. B **761** (2016) 393-398; e-print arXiv: 1511.09316 [hep-lat];
- 4. "The Evolution of Meson Masses in a Strong Magnetic Field"
  M.A. Andreichikov, B.O. Kerbikov, E.V. Luschevskaya, Yu.A. Simonov,
  O.E. Solovjeva,
  JHEP **05** (2017) 007; e-print arXiv: 1610.06887 [hep-ph];
- 5. "Determination of the properties of vector mesons in external magnetic field by Quenched SU(3) Lattice QCD"
  E.V. Luschevskaya, **O.E. Solovjeva**, O.V. Teryaev,
  JHEP **09** (2017) 142; e-print arXiv: 1608.03472 [hep-lat];
- 6. "Tensor polarizability of the vector mesons from $SU(3)$ lattice gauge theory"
  E. V. Luschevskaya, O. V. Teryaev, D. Y. Golubkov, **O. Solovjeva** and R. A. Ishkuvatov,
  JHEP **11** (2018), 186; e-print arXiv: 1811.02344 [hep-lat];
- 7. "Exploring the partonic phase at finite chemical potential within an extended off-shell transport approach "
  P. Moreau, **O. Soloveva**, L. Oliva, T. Song, W. Cassing, E. Bratkovskaya,
  Phys. Rev. C **100** (2019) no.1, 014911; e-print arXiv: 1903.10257 [nucl-th];

- 8. "Transport coefficients for the hot quark-gluon plasma at finite chemical potential μ_B "
O. Soloveva, P. Moreau and E. Bratkovskaya,
Phys. Rev. C **101** (2020) no.4, 045203; e-print arXiv:1911.08547 [nucl-th];
- 9. "Exploring the partonic phase at finite chemical potential in and out-of equilibrium"
O. Soloveva, P. Moreau, L. Oliva, V. Voronyuk, V. Kireyeu, T. Song and E. Bratkovskaya,
Particles **3** (2020) no.1, 178-192; e-print arXiv:2001.05395 [nucl-th];
- 10. "Shear viscosity and electric conductivity of a hot and dense QGP with a chiral phase transition"
O. Soloveva, D. Fuseau, J. Aichelin and E. Bratkovskaya,
Phys. Rev. C **103** (2021) no.5, 054901; e-print arXiv:2011.03505 [nucl-th];
- 11. "Diffusion coefficient matrix of the strongly interacting quark-gluon plasma"
J. A. Fotakis, **O. Soloveva**, C. Greiner, O. Kaczmarek and E. Bratkovskaya,
Phys. Rev. D **104** (2021) no.3, 034014; e-print arXiv: 2102.08140 [hep-ph];
- 12. "QCD at finite chemical potential in and out-of-equilibrium"
O. Soloveva, P. Moreau and E. Bratkovskaya,
Phys. Scripta **96** (2021) no.12, 124067; e-print arXiv:2103.15417 [hep-ph] (2021);
- 13. "Transport properties and equation-of-state of hot and dense QGP matter near the critical end-point in the phenomenological dynamical quasi-particle model",
O. Soloveva, J. Aichelin and E. Bratkovskaya,
Phys. Rev. D, **105** (2022) 5, 054011 e-print arXiv:2108.08561 [hep-ph] (2021);
- 14. "Exploring jet transport coefficients in the strongly interacting quark-gluon plasma", I. Grishmanovskii, T. Song, **O. Soloveva**, C. Greiner and E. Bratkovskaya, Phys. Rev. C, **106** (2022), 014903; e-print arXiv:2204.01561 [nucl-th];
- 15. "Soft gluon emission from heavy quark scattering in strongly interacting quark-gluon plasma", I. Grishmanovskii, T. Song, **O. Soloveva**, Phys. Rev. D, **107** (2023), 036009; e-print arXiv:2210.04010 [nucl-th].
- 16. "Inelastic and elastic parton scatterings in the strongly interacting quark-gluon plasma", I. Grishmanovskii, **O. Soloveva**,
T. Song, C. Greiner and E. Bratkovskaya, Phys. Rev. C under review, e-print arXiv:2308.03105 [hep-ph];
- 17. "Extraction of the microscopic properties of quasi-particles using deep neural networks",
O. Soloveva, A. Palermo, and E. Bratkovskaya,
Phys. Rev. C under review, e-print arXiv:2311.15984[hep-ph];

A complete list of publications can be found under: [INSPIRE: a:"Soloveva, Olga" or a:"Soloveva, O" or a:"Solovjeva, O

Awards and Prizes

2019 Giersch-Excellence-Award for outstanding scientific work during the PhD studies

2019 Best theoretical poster at The XVIII International Conference on Strangeness in Quark Matter (SQM 2019) (young scientist prize sponsored by the Nuclear Physics European Collaboration Committee)

2014 — 2017 Winner of the competition for ITEP young scientists, engineers and technical workers

2015 — 2016 Scholarship for students of the federal state educational institutions of professional education in Russian Federation, Government Decree № 945, Nov 18, 2011.

2015 — 2017 Scholarship for young scientists (senior students, Post-graduate students and candidates of sciences) within the framework of the "Student Research and Education Program of the FAIR-Russia Research Center"

2015 Winner of the competition " NRNU MEPhI best student - 2015"

2016 Edward Lozansky scholarship for students specialized in theoretical physics.

2016 Laureate of the "2^{nd} ITEP premium of the competition in the scientific research (Theoretical works), dedicated to the 70^{th} anniversary of ITEP ". Topic: " Internal structure of mesons in the strong magnetic field".

Selected talks

- Mar 28 - Apr 1, 2022. Mini Workshop "Phase transitions in particle physics", The Galileo Galilei Institute.
  Oral talk title: "Influence of a phase transition on the transport properties of QCD matter" (video).
- Apr 4 - 10,2022. The XXIXth International Conference on Ultra-relativistic Nucleus-Nucleus Collisions (Quark Matter 2022) (online).
  Poster talk title: "Transport properties of the QGP in the dynamical quasi-particle model with a CEP" (pdf).
- Jun 13 - 18, 2022. The 20th International Conference on Strangeness in Quark Matter (SQM 2022), Busan, Republic of Korea.
  Oral talk title: "Exploration of the phase diagram within a transport approach" (pdf).
- Sep 3 - 9, 2021 . Workshop of the Network NA7-Hf-QGP of the European program "STRONG-2020".
  Oral talk title: "Evolution of transport coefficients of the hot and dense QGP along the phase transition".
- May 20 - 25, 2019. the FAIR next generation scientists - 6th Edition Workshop (FAIRNESS2019), Genova, Italy.
  Oral talk title: "Exploring the partonic phase at finite chemical potential within a selfconsistent off-shell transport approach "(pdf).
- Jun 20-25. The XVIII International Conference on Strangeness in Quark Matter (SQM 2019), Bari, Italy.
  Poster title: "Transport coefficients of the hot and dense matter ".
- Jun 18 – 24, 2017. 35^{th} International Symposium on Lattice Field Theory (Lattice 2017), Granada, Spain.
  Oral talk title: "Tetraquark and molecule charmonium states on the lattice".
- Aug 17 – 29, 2017. 6^{th} International Conference on New Frontiers in Physics, Crete, Greece.
  Oral talk title: "Charmed states on the lattice".
- Jul 24 – 30, 2016. 34^{st} International Symposium on Lattice Field Theory (Lattice 2016), Southampton, The United Kingdom.
  Oral talk title: "Lattice simulations of vector mesons in strong magnetic field".

Public talks and Seminars

- 08 May, 2023
  "Dynamical properties of the QGP matter at finite temperatures and chemical potentials", The MUSES Collaboration seminar Series, University of Houston/Illinois/Kent, (online).
- 20 April, 2023
  "Machine Learning for the extraction of the QGP properties",theory seminar at the INFN Sezione di Catania (INFN CT), Catania, Italy.

- 30 March, 2021
  "Shear viscosity and electric conductivity of a hot and dense QGP with a chiral phase transition", Joint UH/UIUC/KSU Weekly journal club, University of Houston, (online).
- 30 March, 2021
  "Shear viscosity and electric conductivity of a hot and dense QGP with a chiral phase transition", Joint UH/UIUC/KSU Weekly journal club, University of Houston, (online).
- 10 December, 2020
  "Transport coefficients of the dense QGP along the chiral phase transition", Transport meeting, ITP Goethe University, Frankfurt am Main, (online).
- 22 June, 2020
  "QGP dynamics at finite chemical potential", IGFAE Theory Seminars at Instituto Galego de Física de Altas Enerxías / Universidade de Santiago de Compostela (online) (video).
- 22 January, 2020
  "Transport properties of the hot and dense QGP", Seminar High Energy Physics, Bielefeld University, Germany.
- 8 July, 2019
  "μ_B-dependence of the QGP dynamics", Palaver at ITP Goethe University, Frankfurt am Main, Germany.
- 14 September, 2017
  "Mesons in strong magnetic field on the lattice", Theory group Literary Seminar at ITEP (Institute for Theoretical and Experimental Physics), NRC Kurchatov Institute, Moscow, Russia.
- 11 December, 2017
  "Light vector mesons in external magnetic field by Quenched SU(3) Lattice QCD", Lattice Seminar at DESY Zeuten, Germany.

Outreach

- 2017
  Public talk "Early Universe and Fundamental Interactions", Cultural and educational project "Kurilka Gutenberga", Moscow, Russia. (Youtube)
- 2015 - 2016
  Organizer of (Quiz: theoretical physics in everyday life) Open faculty days for high school students, NRNU MEPhI, Moscow, Russia.

References

- Prof. Dr. E. Bratkovskaya
- Prof. Dr. J. Aichelin